Assignment 2

Deadline: 22 July 2026 11:30 PM

Submission link: https://forms.gle/6NsNaBMpeYsDZqK38

(Answer any two)

1. Given (-30, -30) and (30, 30) are the corners of the clip rectangle. The endpoints of a line are given as (35, 37) and (-38, -20). Using the Cohen-Sutherland line clipping algorithm, identify whether the lines are accepted, rejected, or partial (partially accepted) only from the out codes. Then determine the clipped line segment. If Cyrus-Beck line clipping algorithm was applied instead of Cohen Sutherland, will it have produced the same output?
2. Given (0, 0) and (400, 300) are the corners of the clip rectangle. The endpoints of a line are given as (−50, 350) and (450, 330). Using the Cyrus-Beck line clipping algorithm, find the clipped line segment.

3. For the following value of t_E and t_L, comment whether the following lines are accepted, rejected or partially clipped:

t_E	t_L	Comments
0.3	0.7	
-0.3	1.2	
-0.3	-0.33	
-0.5	0.5	
1.1	2.4	
0.3	1.3	

4. When the reflected with respect to the line $y = 3x - 8$, the coordinate of the point becomes (3,5). What is the original coordinate of the point?
5. A robot is positioned at (2, 0, 3) with the endpoint of its hand at (2, -2, 1). The robot wants to grab a fruit located at (12, 5, 13). To achieve this, it performs the following transformations sequentially on its hand's endpoint:
   - **Rotation:** At first, the hand's endpoint is rotated 90° counter-clockwise around the X-axis, using the robot's position as the center of rotation.

- **Scaling:** Then, the hand's endpoint is scaled by some factors of Sx, Sy, and Sz. (with respect to robot's position)
- **Translation:** Lastly, the robot itself is translated by (10, 0, 15), bringing the hand's endpoint to the fruit's position so that it can grab the fruit.

Find the values of S that ensure the hand's endpoint aligns with the fruit (the hand's endpoint will be the same as the fruit's location). Show your calculations using the composition of transformations. Also, explain whether the scaling is uniform or non-uniform.

6. A 3D point M is transformed to M' applying shearing along the Y Axis by a shearing factor of (12, 21) with respect to a point (-33, 44, -55). If the **new coordinate M'** is (125, 255, -674). What was the **original coordinate** of **M**?